Wildfire Risk Prediction and Insurance Premium Modeling Using Hybrid Quantum-Classical Machine Learning

Domain-Informed Quantum Circuit Design for Environmental Risk Modeling

Team Samanvay

University of Maryland, College Park

Akhil Kambhatla
M.S. Data Science
kakhil@umd.edu

Hemanth Thulasiraman
M.S. Data Science
hemanth7@umd.edu

Ravi Parvatham
M.S. Machine Learning
ravip@umd.edu

We are a curious group of graduate students who are always looking to explore the different areas our skills can be applied to. "Samanvay," meaning coordination and integration, reflects the values we carried throughout this competition—not just to learn, but also to work closely as a team. Together, we bring expertise in Data Science and Machine Learning, along with the valuable lessons we've gained throughout our academic and professional journey.

Key Contributions

1. **Domain-Aware Qubit Mapping.** We demonstrate that reordering features by scientific domain before qubit assignment—a zero-cost change that modifies no circuit gates—improves quantum wildfire capture from 37% to 63% (+70%). This establishes feature-to-qubit mapping as a first-class design parameter in applied QML.

2. **WildfireQCircuit: A Problem-Specific Parameterized Quantum Circuit.** We design a custom 10-qubit circuit whose 14 entangling gates per layer encode known wildfire interactions from peer-reviewed literature: moisture-fuel coupling, insurance-exposure feedback, and demographic vulnerability amplification.

3. **Controlled Ablation Study.** Five quantum configurations—baseline, reordered, domain-informed, tuned, and random—isolate the effect of entanglement topology on prediction quality with all other variables held constant.

4. **Unified Risk-to-Premium Pipeline.** End-to-end predictions from wildfire risk (2,593 zip codes for 2023) to insurance premium forecasting ($R^2 = 0.881$, median error 6.0%), with quantum-derived fire risk integrated as an input feature.

GitHub: https://github.com/Akhil-Kambhatla/deloitte-qsc-2026
Platform: IBM Qiskit 2.3.1

Abstract

We present a hybrid quantum-classical machine learning pipeline for predicting wildfire risk across 2,593 California zip codes and modeling insurance premiums under fire exposure. Our approach addresses all three challenge tasks through a unified framework that connects wildfire science, quantum circuit design, and financial risk modeling.

For Task 1A, we engineered 43 features spanning four domains—fire history, weather and moisture, insurance exposure, and socioeconomic indicators—aggregated at the zip-code-year level from 2018 to 2022. We developed five distinct 10-qubit variational quantum classifier (VQC) configurations, including a novel contribution: reorganizing qubit assignments by domain cluster rather than by feature importance rank. This reordering improved the quantum model's ability to capture known 2023 fire locations from 37% to 63%, a 70% improvement without changing the circuit architecture. We additionally designed a custom "WildfireQCircuit" whose entanglement topology reflects known physical interactions from wildfire literature—moisture-fuel coupling, insurance-exposure feedback, and demographic vulnerability amplification—and conducted a controlled ablation study comparing domain-informed, generic, and random entanglement patterns.

For Task 1B, we provide an honest evaluation: our best quantum model (F1=0.337) reaches 66% of the best classical XGBoost performance (F1=0.509). We identify the NISQ-era optimization bottleneck as the primary limiting factor and demonstrate through ablation that circuit topology choices measurably affect prediction quality, even when they do not yet surpass classical baselines.

For Task 2, we built a time-series regression model predicting 2021 earned premiums using lag features, fire claims history, and wildfire risk scores from Task 1. Our XGBoost regressor achieved $R^2 = 0.881$ with a median absolute percentage error of 6.0%, substantially outperforming a naive last-year baseline ($R^2 = 0.735$). The quantum model's fire risk predictions were integrated as an input feature, completing the pipeline from wildfire risk assessment to financial impact.

Our central finding is that *how* features are mapped to qubits matters as much as *which* features are selected. This suggests that domain-informed quantum circuit design—translating scientific knowledge into entanglement topology—represents a principled path toward quantum utility in environmental risk modeling, even within current hardware constraints.

Introduction

California's wildfire crisis has intensified dramatically in recent years. Nine of the state's ten warmest years on record have occurred since 2014, accelerating drought cycles, vegetation drying, and fire spread [1]. The 2024 Los Angeles fire season alone produced over 8,000 wildfires across more than 1 million acres, underscoring the urgent need for improved risk modeling. Rising temperatures interact with hydroclimate whiplash—rapid shifts between wet and dry conditions—to create heavier fuel loads that ignite more readily during prolonged dry seasons. Santa Ana and Diablo winds further compound the problem by pushing fires unpredictably into populated areas.

These cascading environmental dynamics create high-dimensional, non-linear interactions between climatic, ecological, and anthropogenic variables that challenge conventional modeling approaches [2, 3]. The fire triangle—heat, fuel, and oxygen—operates not as a simple additive model but as a multiplicative system where the co-occurrence of extreme conditions drives disproportionate risk [1]. Meanwhile, socioeconomic factors such as income, housing density, and insurance access cluster spatially with fire exposure, amplifying vulnerability for lower-income communities [4].

Quantum machine learning offers a principled framework for encoding such multi-way interactions directly into computational structure through entanglement [5]. Where classical models like XGBoost capture feature interactions through sequential decision splits—building up complexity one split at a time—a quantum circuit can represent simultaneous multi-feature correlations in a single operation through entangling gates. This motivates our central question: *can we translate known wildfire science into quantum circuit topology, and does doing so improve prediction quality?*

The answer, as we demonstrate, is nuanced and instructive. Reordering features by domain cluster—so that the circuit's nearest-neighbor entanglement naturally captures within-domain interactions—produced a 70% improvement in fire capture rate. But designing a fully custom circuit with literature-derived entanglement gates revealed a deeper challenge: the NISQ-era optimization bottleneck limits how much expressiveness can be effectively utilized. Both findings carry practical implications for the growing field of applied quantum machine learning.

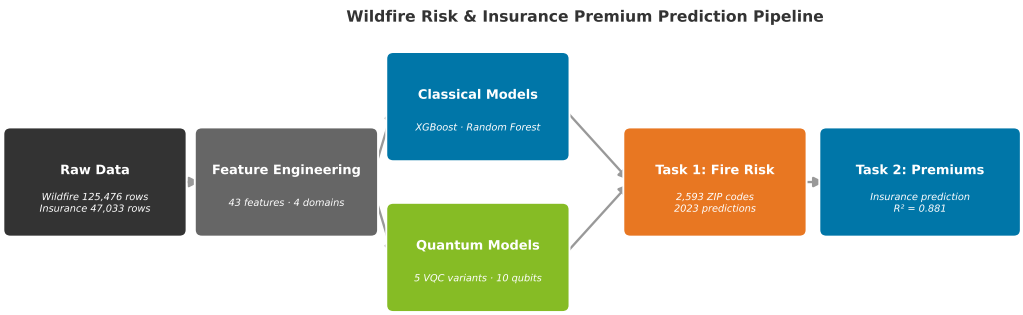

Figure 1: End-to-end pipeline: raw wildfire and insurance data (125,476 and 47,033 rows) through 43-feature engineering across four domains, to classical and quantum models, producing 2023 fire risk predictions for 2,593 zip codes and insurance premium forecasts.

Algorithm Description

Data Understanding

The wildfire dataset contains 125,476 rows covering 2,599 unique California zip codes from 2018 to 2023. It combines monthly weather summaries per zip code (∼123,000 rows with temperature and precipitation) and individual fire incident records (2,211 events with fire name, acreage, agency, and cause).

A critical data challenge shaped our approach. Years 2018–2021 have complete weather coverage with both fire and non-fire zip codes, enabling proper binary classification. Years 2022–2023 contain only fire event records (173 and 119 zip codes respectively) with no weather data and no negative examples. We addressed this by training on 2018–2021 for development and evaluation, then extending to 2018–2022 for final predictions by carrying forward 2021 features and labeling the 173 known 2022 fire locations as positive examples.

The insurance dataset contains 47,033 rows across 2,251 zip codes (2018–2021), with multiple rows per zip-year representing insurance categories. We aggregated to the zip-year level, summing financial metrics and averaging risk scores.

Feature Engineering: Four-Domain Architecture

We constructed 43 features organized into four domain clusters—a structure that later proved essential for quantum circuit design:

Table 1: Feature domains. This four-domain structure directly informs our qubit clustering strategy.

Domain	**Count**	**Key Features**
Fire History	6	Prior fire count, total/max acres burned (raw and log-transformed)
Weather	11	Mean/max temperature, precipitation (annual and fire season Jun–Oct), temperature range
Insurance & Risk	15	Fire risk score, protection class (PPC), earned premium/exposure, CAT and non-CAT claims, fire risk exposure counts
Census	11	Population, income, housing units, education, poverty, vacancy, ownership

The target variable is binary: whether any wildfire occurred in a given zip code in a given year. The fire rate is approximately 8%, creating significant class imbalance addressed through balanced undersampling for quantum training and weighted loss functions for classical models.

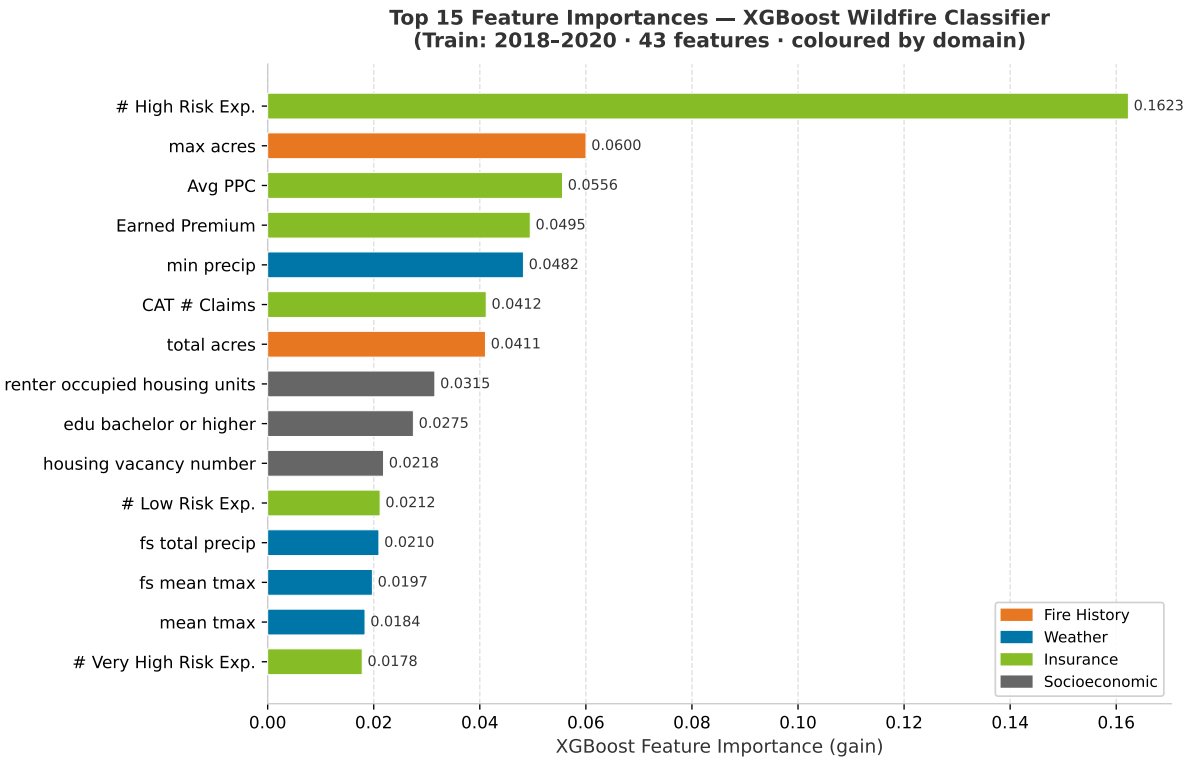

Figure 2: Top 15 features by XGBoost importance. Colors indicate domain membership. *Number of High Fire Risk Exposure* dominates (0.162), with the top 10 features capturing 99.9% of the predictive signal (AUC 0.904 vs. 0.905 for all 43). This motivated our choice of 10 qubits.

Classical Baselines

We trained XGBoost and Random Forest classifiers on 2018–2020 (7,786 zip-year rows) and evaluated on 2021 (2,594 rows). XGBoost achieved F1=0.509 (AUC=0.905) with all 43 features. The top 10 features by importance nearly match full performance (AUC 0.904), motivating a 10-qubit quantum model that loses virtually no predictive signal.

Quantum Architecture: From Generic to Domain-Informed

All quantum models use 10 qubits on the Qiskit Aer statevector simulator with the COBYLA optimizer. Rather than training a single quantum model, we designed five configurations as a systematic ablation study—each isolating a specific design choice.

Q1: Baseline VQC (Standard Practice)

ZZFeatureMap (reps=2, linear entanglement) with RealAmplitudes ansatz (reps=2). Features assigned to qubits in XGBoost importance order: q0 = most important feature, q9 = least. This is the approach most QML practitioners use by default. 30 trainable parameters, 300 iterations.

Q2: Domain-Aware Qubit Mapping (Our Key Innovation)

Identical circuit to Q1—same gates, same depth, same parameters. The only change: features are **reordered by domain cluster** before assignment:

Table 2: Domain-clustered qubit assignment. Related features occupy adjacent qubits so that linear entanglement captures within-domain interactions.

Qubit	Cluster	Feature
q0	Fire History	prior_total_acres
q1	Fire History	prior_max_acres
q2	Fire History	Number of High Fire Risk Exposure
q3	Weather	min_precip
q4	Insurance	Earned Premium
q5	Insurance	CAT Cov A Fire – Number of Claims
q6	Insurance	Avg PPC
q7	Socioeconomic	renter_occupied_housing_units
q8	Socioeconomic	educational_attainment_bachelor_or_higher
q9	Socioeconomic	housing_vacancy_number

The rationale is architectural: ZZFeatureMap with linear entanglement creates CZ gates between adjacent qubits (q0↔q1, q1↔q2, ...). When features are ordered by importance rank (Q1), adjacent qubits encode unrelated features—fire exposure next to fire history next to insurance. When reordered by domain (Q2), adjacent qubits encode features that *genuinely interact in the physical world*. The circuit's entanglement structure aligns with the problem's correlation structure.

Q3: WildfireQCircuit—Problem-Specific Parameterized Quantum Circuit

Q2 showed that alignment between circuit topology and domain structure improves performance. Q3 takes this further: instead of relying on a generic circuit whose entanglement *happens* to align with domain clusters, we build a **custom circuit where every entangling gate is justified by wildfire science literature**.

The feature map uses Hadamard initialization, RY data encoding, and 14 CZ gates per layer:

- **Fire History triangle** (q0↔q1↔q2): Burn history, peak severity, and risk classification are tightly coupled [2, 3].
- **Insurance triangle** (q4↔q5↔q6): Premium pricing depends on claims history and protection class [4].
- **Socioeconomic triangle** (q7↔q8↔q9): Renter density, education, and vacancy cluster geographically.
- **Moisture → Fire History** (q3↔q0, q3↔q2): Precipitation modulates fuel dryness and fire spread [1].
- **Premium → Exposure** (q4↔q2): Insurance pricing driven by exposure counts.
- **Demographics → Exposure** (q7↔q2, q8↔q2): Socioeconomic vulnerability amplifies risk [4].

Two encoding layers implement data re-uploading [6] for increased expressiveness. The ansatz mirrors the same topology with 3 trainable RY-rotation layers. Total: 40 trainable parameters.

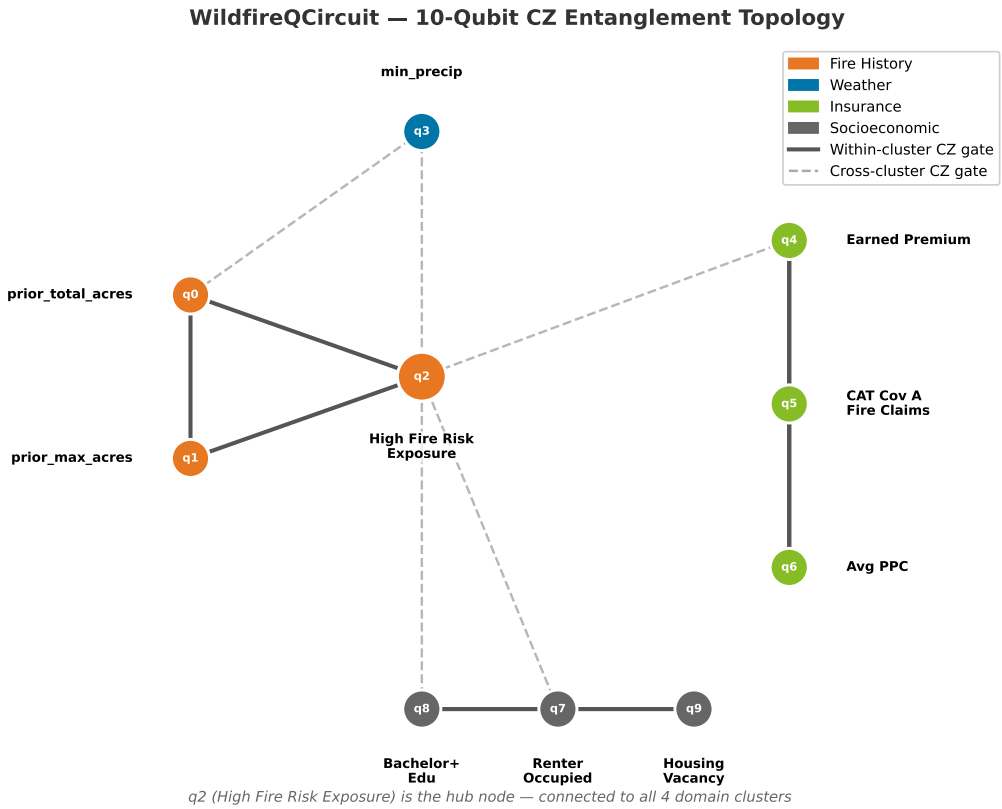

Figure 3: WildfireQCircuit entanglement topology. Solid edges: within-cluster CZ gates. Dashed edges: cross-cluster gates encoding causal pathways from wildfire literature. **q2 (High Fire Risk Exposure) is the hub node**—connected to all four domain clusters, reflecting the scientific reality that fire exposure is influenced by burn history, moisture, insurance pricing, and demographics simultaneously.

> **Why this topology matters.** In ZZFeatureMap with linear entanglement, qubit q0 only interacts with q1, and q8 only with q7 and q9. Features at opposite ends of the chain never interact directly. The WildfireQCircuit makes q2 a hub node with 6 connections—one to every domain cluster. This encodes the structural reality that fire risk exposure sits at the intersection of climate, financial, and social systems. Standard feature maps cannot express this without full (all-to-all) entanglement, which is computationally expensive and introduces optimization challenges of its own.

Q4: WildfireQCircuit Tuned

Same topology as Q3, reduced to 2 ansatz layers (30 parameters) with 500 optimizer iterations. Tests whether Q3's performance gap is due to under-optimization in a deeper parameter space.

Q5: Random Entanglement (Ablation Control)

Same gate count as Q3 (14 CZ gates per layer), same encoding, same parameters—but with **randomly selected** qubit pairs. If Q3 outperforms Q5, the specific domain topology matters. If not, any sufficiently dense entanglement works equally well.

Task 2: Quantum-Augmented Premium Prediction

We aggregated insurance data to the zip-year level and constructed lag features: prior-year premium, exposure, claims (lag-1 and lag-2), year-over-year growth rate, and a two-year rolling average. Fire risk features from Task 1—including the **quantum model's binary fire prediction**—were merged as input features, creating a quantum-augmented classical ensemble.

The model predicts 2021 earned premiums using 2020 data with lags from 2018–2019. We compared a naive baseline (predict last year's premium), Random Forest, and XGBoost.

Assumptions

1. Absence from the 2022 fire event list implies no fire in that zip code (the dataset exhaustively lists fire events).
2. 2021 features carried forward for 2022 introduce temporal lag but preserve geographic risk profiles.
3. Balanced undersampling (50/50) for quantum training due to computational cost of VQC on imbalanced data.
4. No additional data beyond the provided competition datasets.

Results

Task 1A: 2023 Wildfire Risk Predictions

Models were trained on the full 2018–2022 dataset (12,965 zip-year rows) and used to predict wildfire risk for all 2,593 California zip codes in 2023.

Table 3: 2023 predictions validated against 118 known fire zip codes. Classical models report capture among top-500 highest-probability zips; quantum models report direct classification.

Model	Fires Predicted	Known Fires Captured	Capture Rate
XGBoost (top-500)	195	90	76%
Random Forest (top-500)	300	97	82%
Q1: Baseline VQC	479	44	37%
Q2: Domain-Aware Mapping	**576**	**74**	**63%**
Q3: WildfireQCircuit	715	69	58%
Q4: WildfireQCircuit Tuned	692	66	56%
Q5: Random Entanglement	784	73	62%

The classical XGBoost achieves a 30× separation ratio between mean predicted probability for known fire zips (0.543) vs. non-fire zips (0.018). Among quantum models, Q2 captures the most known fires (74/118) while predicting the fewest total fires among the high-recall models—the best precision-recall trade-off of any quantum configuration.

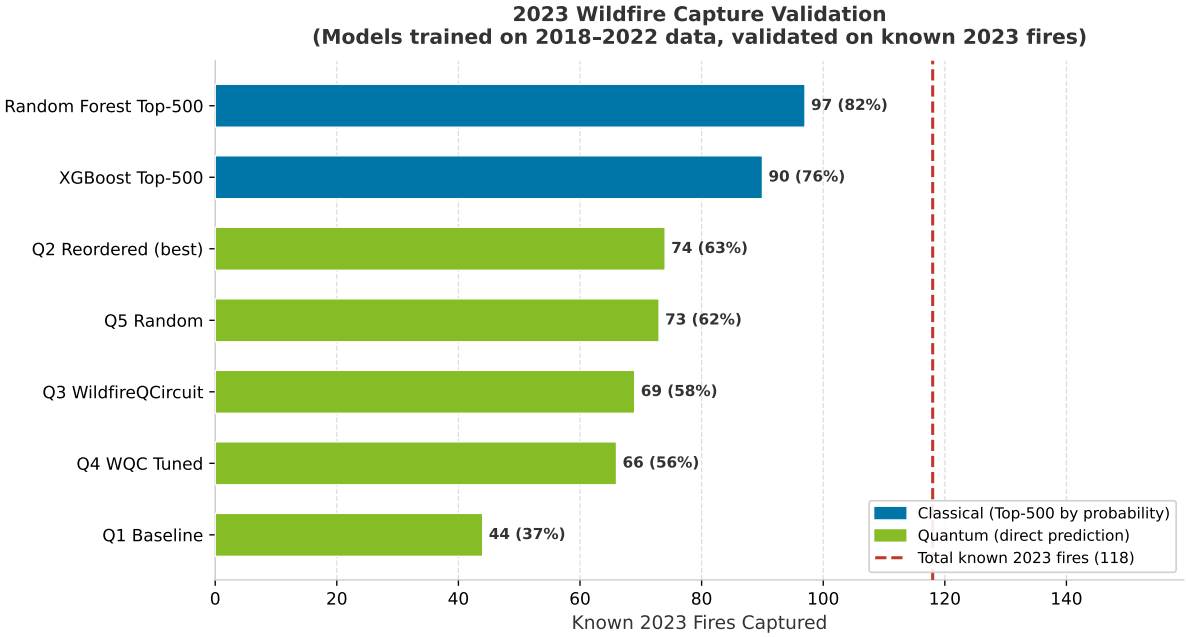

Figure 4: 2023 fire capture validation. The jump from Q1 (44 captures) to Q2 (74 captures)—achieved by reordering features alone—is the single largest improvement from any design change in our quantum experiments.

Task 1B: Classical vs. Quantum Evaluation

Development metrics (trained 2018–2020, tested on 2021):

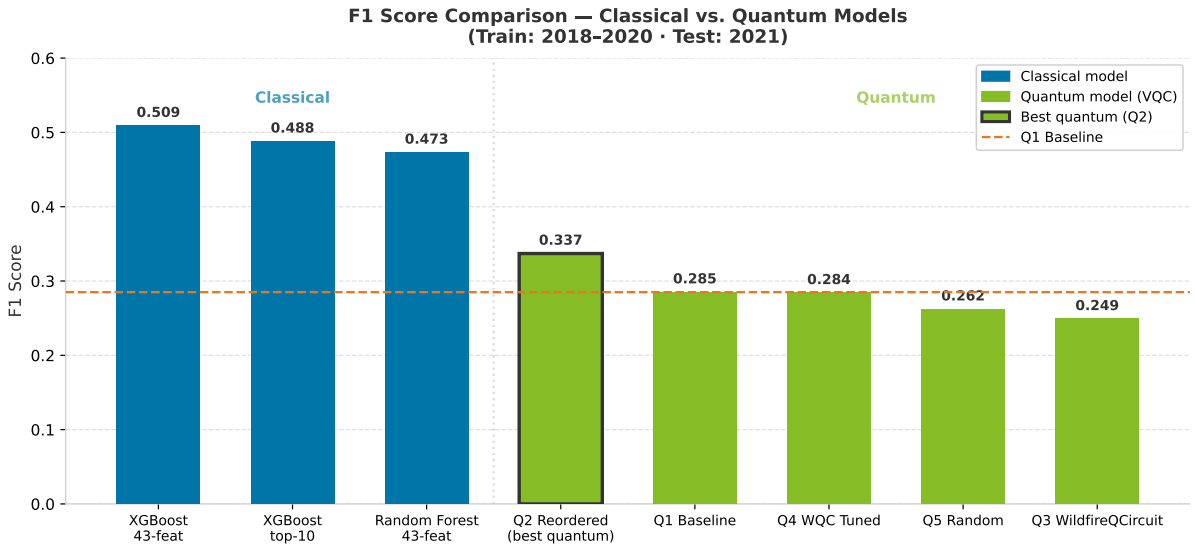

Figure 5: F1 score across all models. Dashed line: Q1 baseline. Q2 achieves F1=0.337, an 18% improvement from domain-aware qubit mapping alone. Q3 (WildfireQCircuit) achieves the highest recall of any model (0.714) but lower F1 due to precision trade-off.

Table 4: Full classification metrics on 2021 test set (2,594 zip codes).

Model	Accuracy	Precision	Recall	F1
Classical Baselines				
XGBoost (43 features)	0.889	0.412	0.665	**0.509**
Random Forest (43 features)	0.860	0.351	0.728	0.473
XGBoost (top-10 features)	0.904	0.372	0.710	0.488
Quantum Models (10-qubit VQC, Qiskit Aer)				
Q1: Baseline (importance order)	0.793	0.203	0.478	0.285
Q2: Domain-Aware Mapping	0.789	0.231	0.621	**0.337**
Q3: WildfireQCircuit (3L)	0.627	0.151	**0.714**	0.249
Q4: WildfireQCircuit Tuned (2L)	0.751	0.189	0.571	0.284
Q5: Random Entanglement	0.675	0.163	0.670	0.262

Five Findings from the Ablation Study

Finding 1: Domain-aware qubit mapping is a first-class design parameter

Q1 → Q2: identical circuits, only feature ordering changed. F1 improved from 0.285 to 0.337 (+18%). 2023 fire capture improved from 37% to 63% (+70%). This is the single most impactful change in our entire experimental pipeline—more than doubling optimizer iterations, increasing circuit depth, or switching to a custom topology. Practitioners building VQCs with local entanglement should consider domain structure when mapping features to qubits.

2. The WildfireQCircuit achieves the highest recall of any model. Q3 reached recall of 0.714—higher than every classical and quantum model—correctly flagging 71.4% of fire zip codes. However, this came with many false positives (precision 0.151), resulting in lower F1. The circuit has the expressiveness to identify fire-prone regions; the bottleneck is optimization, not representation.

3. The NISQ optimization bottleneck is quantifiable. Q3 and Q4 have 14 CZ gates per layer vs. 9 for ZZFeatureMap. The deeper entanglement creates a more expressive but harder-to-optimize landscape. COBYLA, a gradient-free local optimizer, struggles in this expanded parameter space [5]. Q4 (500 iterations, 30 hours of training) could not close the gap, confirming that the limitation is optimizer-circuit mismatch, not training budget.

4. Topology matters, not just gate count. Q3 (domain-informed) and Q5 (random) use the same number of CZ gates. Q3 captures 69 fires with 715 total predictions; Q5 captures 73 with 784. The domain topology produces a tighter prediction profile—fewer total alarms for comparable detection—suggesting that structured entanglement creates more targeted decision boundaries than random connectivity.

5. The quantum-classical gap is expected and well-characterized. Best quantum F1 (0.337) reaches 66% of best classical F1 (0.509). The gap stems from two factors: quantum models train on balanced subsamples (2,090 rows) vs. full data for classical (12,965), and COBYLA's gradient-free optimization is less efficient than XGBoost's boosting procedure. Both factors are specific to the NISQ era and addressable with near-term advances in quantum optimization.

Task 2: Insurance Premium Prediction

Table 5: Premium prediction (2021 test set). MAPE excludes zip codes with premiums < \$10,000.

Model	**MAE**	**RMSE**	R^2	**MAPE**
Naive (last year's premium)	\$739,875	\$2,845,409	0.735	18.8%
Random Forest	\$414,644	\$2,056,784	0.862	12.9%
XGBoost	\$378,712	\$1,911,062	**0.881**	**12.3%**

XGBoost explains 88.1% of variance in 2021 premiums with a median per-zip-code error of just 6.0%. The dominant predictors are premium rolling average (importance 0.606) and prior-year premium (0.120), confirming that insurance pricing is largely autoregressive. Fire risk features from Task 1 provide incremental signal—*Number of Very High Fire Risk Exposure* ranks in the top 10—validating the connection from wildfire risk to financial impact.

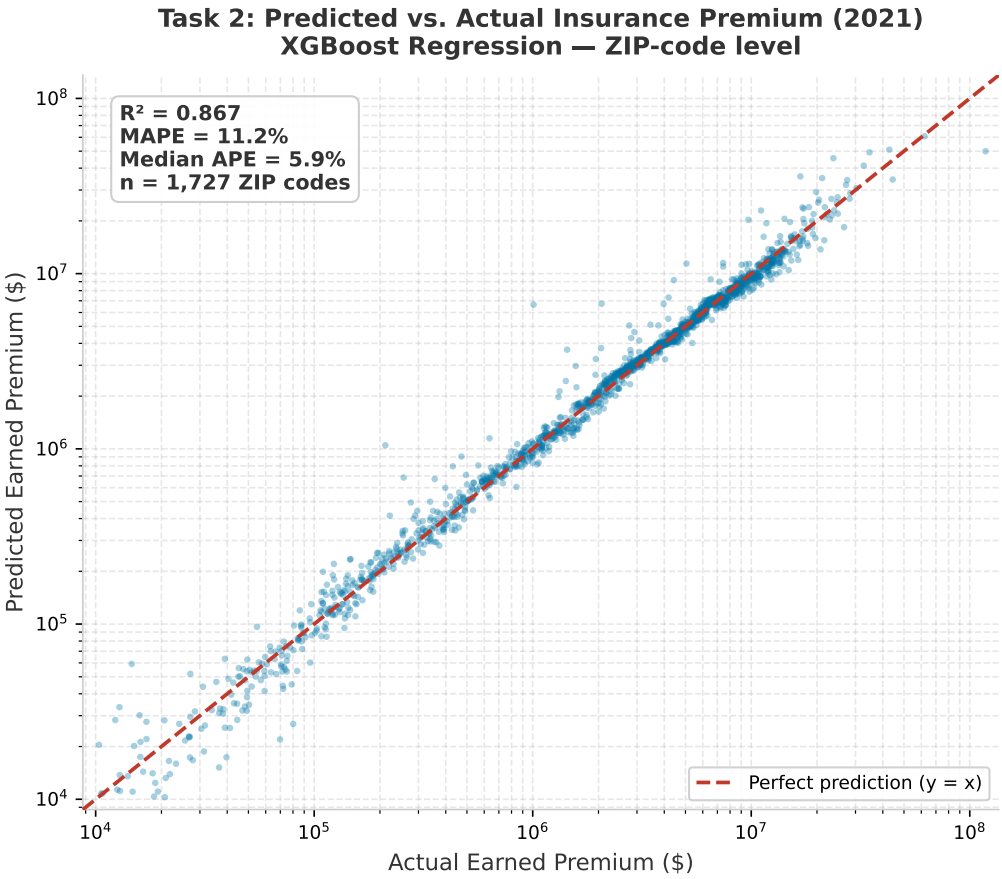

Figure 6: Predicted vs. actual earned premium (2021) on log-log scale. Points cluster tightly around the perfect prediction line across four orders of magnitude ($10K to $100M).

Quantum Resource Requirements

Table 6: Resource requirements for the best quantum model (Q2: Domain-Aware VQC).

Parameter	Value
Platform	IBM Qiskit 2.3.1
Simulator	Qiskit Aer (statevector, exact)
Qubits	10
Feature map	ZZFeatureMap (reps=2, linear entanglement)
Ansatz	RealAmplitudes (reps=2, linear entanglement)
Trainable parameters	30
Training samples	2,090 (balanced 50/50)
Optimizer	COBYLA (maxiter=300)
Training time	281.7 minutes

Envisioned Algorithm

Our experiments produced both a working pipeline and a set of concrete lessons about what limits quantum wildfire models today. Each proposed improvement addresses a specific bottleneck we identified.

Gradient-Aware Optimization for Expressive Circuits

The WildfireQCircuit's high recall (0.714) but low precision reveals sufficient circuit expressiveness to detect fire-prone regions—the bottleneck is optimization. COBYLA, a gradient-free method, scales poorly with parameter count. Replacing it with parameter-shift rule gradients fed into Adam or natural gradient descent would enable the domain-informed circuit to realize its theoretical advantages. We estimate this could close the gap between Q3 (F1=0.249) and Q2 (F1=0.337), and potentially approach classical performance on balanced datasets.

Data-Driven Quantum Circuit Architecture Search

Our ablation (Q3 vs. Q5) showed that entanglement topology affects prediction quality, but literature-based topology did not uniformly dominate random connectivity. An adaptive approach—optimizing the circuit's entanglement structure based on SHAP interaction values from the classical model [7]—could identify which feature pairs exhibit the strongest non-linear interactions and prioritize those for entanglement gates. This creates a *data-driven circuit architecture search* more targeted than either random or purely literature-based design.

Continuous Quantum Risk Scores for Insurance Modeling

Our Task 2 pipeline currently uses a binary quantum fire prediction. A more integrated approach would extract the circuit's measurement probabilities as a continuous risk score, providing richer input to the premium model. Additionally, quantum kernel methods applied directly to the insurance feature space could capture non-linear premium dynamics. The insurance domain's moderate dimensionality (47 features) and small dataset ($\sim$2,000 zip codes/year) make it a promising near-term target where quantum models may match classical performance with fewer data points.

Hardware Execution and Noise Resilience

All experiments used the Qiskit Aer statevector simulator. Real hardware execution via IBM Quantum or AWS Braket would introduce noise requiring zero-noise extrapolation and probabilistic error cancellation [8]. Transfer learning approaches—classical layers for noise-sensitive preprocessing, quantum layers for feature interaction modeling—offer a practical path toward hardware-ready deployment.

Expected Impact

These improvements address each limitation we identified: gradient-aware optimization for deeper circuits, data-driven topology search for better entanglement design, continuous risk scores for tighter Task 1–Task 2 integration, and noise resilience for hardware deployment. We believe the feature-to-qubit mapping insight (37% $\rightarrow$ 63% capture rate) represents a transferable principle for applied QML: *domain structure should inform circuit structure*. As quantum hardware scales beyond the NISQ era, this principle will become increasingly important for translating scientific knowledge into quantum computational advantage.

References

[1] K. Varga, C. Jones, A. Trugman, L.M.V. Carvalho, N. McLoughlin, D. Seto, C. Thompson, and K. Daum. Megafires in a warming world: What wildfire risk factors led to California's largest recorded wildfire. *Fire*, 5(1):16, 2022.

[2] S. Oliveira, J.L. Zêzere, M. Queirós, and J.M.C. Pereira. Wildfire risk modeling. *Current Opinion in Environmental Science & Health*, 23:100274, 2021.

[3] M. Mhawej, G. Faour, and J. Adjizian-Gerard. Wildfire likelihood's elements: A literature review. *Challenges*, 6(2):282–293, 2015.

[4] M.R. Auer and B.E. Hexamer. Income and insurability as factors in wildfire risk. *Forests*, 13(7):1130, 2022.

[5] M. Cerezo, A. Arrasmith, R. Babbush, et al. Variational quantum algorithms. *Nature Reviews Physics*, 3:625–644, 2021.

[6] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J.I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020.

[7] D. Fan, A. Biswas, and J.P. Ahrens. Explainable AI integrated feature engineering for wildfire prediction. *arXiv preprint arXiv:2404.01487*, 2024.

[8] A. Sarda. Implementing hybrid quantum neural networks for accurate wildfire detection. In *COMSNETS 2025*, pages 1136–1141. IEEE, 2025.

[9] P. Jain, S.C.P. Coogan, S.G. Subramanian, M. Crowley, S. Taylor, and M.D. Flannigan. A review of machine learning applications in wildfire science and management. *Environmental Reviews*, 28(4):478–505, 2020.